

datasheet-rag — Case Study

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Problem

Engineering answers live in datasheets — dense, table-bound, and conditions-qualified (“ $I_Q = 25\ \mu\text{A}$ at $3.3\ \text{V}$, $-40\ ^\circ\text{C}$ ”), where part numbers punish fuzzy matching (an LM358 is not an LM359). Naive retrieval-augmented generation (RAG) fails on exactly these properties. I built a RAG system over 90 semiconductor datasheets that treats those properties as the design brief, and — just as importantly — **measured every design choice instead of assuming it.**

Approach

Pipeline. Table-aware PDF parsing (Docling) keeps tables as first-class chunks; dense retrieval (nomic-embed + Chroma) is reranked by a cross-encoder; a 4B model generates grounded answers with inline citations and an explicit refusal path. Everything runs on open weights with a local judge — no closed-API dependency.

Evaluation first. Before optimizing anything, I built a 100-question golden set with every value and evidence chunk verified against the index, scored by a *separate* judge model (to avoid self-preference bias), plus a synthetic generator and judge-vs-human agreement tooling. Measurement, then optimization.

Results — where the project earns its keep

The interesting findings are the ones that contradicted the obvious move.

1. Hybrid retrieval didn't help; reranking did. The textbook fix for part-number queries is to add BM25. I measured it: the BM25+RRF hybrid *underperformed* dense alone. Root cause — dense already routed to the correct document 100 % of the time, and my own per-chunk part-number prefix left BM25 no within-document discrimination. The real residual failure was chunk ranking *within* the right document, which a cross-encoder reranker fixed:

Retriever	hit@8	recall@8	MCU hit@8
dense	0.92	0.90	0.75
naive BM25+RRF hybrid	0.86	0.84	0.62
dense + cross-encoder rerank	0.97	0.95	0.94

2. Fine-tuning didn't help either; it hurt. I ran a QLoRA study (Qwen3-4B, 4 arms over frozen contexts so the model was the only variable). Retrieval dominated: closed-book correctness ~2/5 vs RAG ~4.6/5. Worse, fine-tuning on synthetic Q/A *collapsed refusal calibration* — the model refused only 3/9 unanswerable questions vs the base model's 8/9 — because the training set contained no “I don't know” examples and taught the model to always answer. The evidence said: don't ship the fine-tune. So I didn't.

Final system: dense + rerank — hit@8 0.97, correctness 4.39/5, refusal calibration 9/9.

What it demonstrates

Applied LLM/RAG engineering end-to-end — document ingestion, retrieval, evaluation methodology, controlled ablation, parameter-efficient fine-tuning — on bleeding-edge infrastructure (CUDA 13 / Blackwell), with a discipline that matters more than any single technique: **measure, let the evidence decide, and report what doesn't work.** Twice the obvious approach failed and I shipped the configuration the data supported instead. That judgment — including declining to ship a fine-tuned model because it was measurably worse — is the point.

Source, full ablations, and reproducible scripts: github.com/palginpav/datasheet-rag